



DEPARTMENT OF APEX INSTITUTE OF TECHNOLOGY

PROJECT PROPOSAL

1. Project Title: -

Automated Resume Screening Using NLP and Bias Analysis.

2. Project Scope: - (Max 500 words)

1. **Project Overview** The primary objective of this project is to develop an intelligent, AI-driven application designed to automate the initial screening phase of the recruitment process. By leveraging **Natural Language Processing (NLP)** and **Machine Learning (ML)** techniques, the system will ingest, analyze, and rank job applications against specific job descriptions. The core philosophy of this project is twofold: to drastically improve recruitment efficiency by automating high-volume tasks and to strictly enforce **Ethical AI** principles by detecting and mitigating algorithmic bias.
2. **Functional Scope** The technical scope of the project encompasses the end-to-end processing of candidate data:
 - **Data Ingestion & Parsing:** The system will accept resumes in various formats (PDF, DOCX) and use NLP parsing to extract unstructured text. Key entities such as candidate names, educational background, technical skills, and work experience will be identified and structured for analysis.
 - **Semantic Matching & Ranking:** Unlike simple keyword matching, the system will utilize advanced ML models (e.g., word embeddings or transformer models) to understand the semantic context of a resume. It will compare candidate profiles against job descriptions to calculate a "relevance score," effectively shortlisting the most suitable candidates.
 - **Automation of Workflow:** The project includes the creation of a pipeline that automatically categorizes applications, reducing the manual workload on HR teams and minimizing human error inherent in fatigue-driven reviews.
3. **Bias Detection and Ethical AI Scope** A critical and defining component of this project scope is the integration of a **Bias Analysis Module**. Recognizing that automated systems often inherit prejudices from historical hiring data, this project explicitly expands its scope to include:
 - **Fairness Evaluation:** The system will be tested against fairness metrics to ensure it does not discriminate based on gender, ethnicity, or other protected characteristics.
 - **Bias Mitigation Techniques:** The project will implement algorithms designed to "blind" the model to irrelevant demographic data or re-weight features to ensure equitable scoring.
 - **Auditing & Explainability:** The scope includes developing mechanisms to audit decision-making pathways, ensuring that the reasons for shortlisting a candidate are transparent and based solely on merit and skill alignment.

4. Project Outcomes Successful delivery of this project will result in a deployment-ready software tool that reduces time-to-hire and administrative costs. More importantly, it will deliver a **fair recruitment tool** that upholds equal opportunity standards, providing the organization with a diverse and high-quality talent pool while protecting against the reputational and legal risks associated with biased hiring practices.

3. Requirements: -

➤ Hardware Requirements

☐ Processor (CPU) & GPU:

- Minimum: Intel Core i5 (10th Gen) or AMD Ryzen 5.
- *Recommended:* NVIDIA GPU (GTX 1650 or higher) or Google Colab (Cloud GPU) is highly recommended for faster training of NLP models and handling deep learning tasks.

☐ Random Access Memory (RAM):

- Minimum: 8 GB.
- *Recommended:* 16 GB is preferred to efficiently load large NLP models (like BERT or Spacy pipelines) and process datasets without system lag.

☐ Storage:

- Minimum: 256 GB SSD (Solid State Drive).
- *Why:* SSDs provide the necessary speed for reading large numbers of resume files and loading heavy machine learning models compared to traditional HDDs.

➤ Software Requirements:

☐ Operating System:

- Windows 10/11 (64-bit), macOS, or Linux (Ubuntu).

☐ Programming Language:

- **Python (Version 3.8 or higher):** Python is the industry standard for AI and NLP projects due to its extensive library support.

☐ Libraries & Frameworks:

- **NLP & ML:** TensorFlow or PyTorch (for Deep Learning), Spacy or NLTK (for text processing), Scikit-learn (for bias metrics).
- **Data Handling:** Pandas and NumPy.
- **Interface:** Flask, Django, or Streamlit (to build the web interface for uploading resumes).
- **IDE:** VS Code, PyCharm, or Jupyter Notebook.

STUDENTS DETAILS

Name	UID	Signature
BEDANT SINGHANIA	23BDA70122	
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APPROVAL AND AUTHORITY TO PROCEED

We approve the project as described above, and authorize the team to proceed.

Name	Title	Signature (With Date)